

INSTITUTO POLITÉCNICO NACIONAL
ESCUELA SUPERIOR DE CÓMPUTO



Selected Topics in Cryptography

Lab 06: CTR and RSA-OAEP *October 20, 2026*

1 Schoolbook RSA with chinese remainder theorem

Do the following exercises by your own. You can use C/C++, C#, Java or Python to develop your source code, but please use only one of this languages to develop all the exercises. At the end of the session upload to Microsoft Teams screenshots of your source code and screenshots of your programs running. To develop these exercises you **must NOT USE any cryptographic library to implement RSA**.

1. Implement a function to generate a key pair for schoolbook RSA with CRT whenever it is required. Notice that the private key must a tuple: (p, q, n, d, d_p, d_q), where $|p| = |q| \geq 4$ bits; $n = p * q$, and d is the private exponent; $d_p = d \bmod p - 1$ and $d_q = d \bmod q - 1$. The public key must be (e, n), e must be randomly generated, **DO NOT USE** $e = 3$ **or** $e = 2^{16+1}$ To generate the prime numbers use a function within the programming language you have chosen. Do not implement a function of your own creation to do this. Print the private key and the public key.

2. Implement the pseudocode of Algorithm 1 in a function.

Algorithm 1: Square and Multiply(x, H, n)

Input : $H = (b_t, \ldots, b_1, b_0)_2$ binary representation of H; $n \geq 0$ and $0 \leq x < n$
Output: $x^H \bmod n$

```
r ← x ;
for i ← t − 1 to 0 do
    r ← r^2 mod n ;
    if b_i == 1 then
        r ← r * x mod n
    end
end
return r
```

3. Implement a function to encipher an integer $0 < m < n$ using schoolbook RSA. The function must receive as input m and the public key (e, n). The value of m and the public key must be given by the user. Your function must return the ciphertext c. Use your implementation of Algorithm 1 to encipher m. Also print the value of c

4. Implement a function to decipher an integer $0 < c < n$ using schoolbook RSA and CRT. The function must receive as input c and the private key (p, q, n, d, d_p, d_q) . The value of c and the private key must be given by the user. Your function must return the ciphertext c. Also you must print each value that you compute when you are using the chinese remainder theorem.

5. Your program must be able to do separately each one of the procedures: key pair generation, encipher a message and decipher the corresponding ciphertext .

6. Test each function and generate 5 different key pairs, encipher 10 different messages, decipher each ciphertext that you generated.

2 RSA-OAEP with CRT

Do the following exercises with your project team. Avoid to use fixed data such as filenames. To develop these exercises you **must USE a cryptographic library to implement RSA OAEP with CRT**.

1. Implement a function to generate a key pair for RSA-OAEP with CRT. The private key and the public key must go to different textfiles. **You must store each key using base64**.

2. Generate a random key K of 128 bits for AES and print it in base64. Use it to encipher a file (with any format).

3. Implement a function to encipher K using a public key for RSA-OAEP, store the enciphered key K in a file in base64. The user must be able to choose the filename.

4. Implement a function to decipher the enciphered K using the corresponding private key for RSA-OAEP with CRT. Print the deciphered K in base64.

5. Test your functions as follows:

   - One student will play the role of Alice and the other one will play the role of Bob.
   - Alice will generate her key pair for RSA OAEP with CRT. Alice will share the file in base64 of her public key with Bob.
   - Bob will generate a random key K of 128 bits for AES. Then he will encipher a file m using K and AES. Then he will encipher this key K with RSA-OAEP using the public key of Alice and store the enciphered key in a file in base64. Bob will share the enciphered key and the ciphertext with Alice.
   - Alice will decipher the enciphered key with her private key using RSA-OAEP with CRT. Once she recover K, she must decipher the ciphertext with AES.

3 Questions

Read the standard of NIST SP 800-56BR2 to answer the following questions:

1. What is the title of NIST SP 800-56BR2?

2. Read Section 6.2.2 of NIST SP 800-56BR2. Which are the three formats for the private key of RSA OAEP?

3. In the case of CRT format for private key. How many values does the key have? What is each value?

4. Read section 6.3 of NIST SP 800-56BR2and find the security levels for each modulus length of RSA-OAEP. Which length did you use in your implementation? Which line of source code let you establish the length of the modulus?

5. Read section **7.1.2.3 Decryption with the private key in the CRT format** of NIST SP 800-56BR2 and explain the process to decrypt using a toy example.

6. Read section **9.2 KTS-OAEP: Key transport using RSA-OAEP** What is KTS-OAEP used for? Describe in your own words each step.

4 Products

Every student must write her/his own report including the following information:

1. Personal information, date of the lab session and the topic that we are studying in this lab session.

2. Source code for each function of Section 1. List of the generated key pairs, plaintexts and ciphertexts using to test your functions.

3. Answers to questions of Section 3.